

Food Traceability and Bioanalysis

Science

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Food Traceability and Bioanalysis (A13090)

Short Title: Food Traceability Bioanalysis
Faculty Science and Computing
Department Science
Cluster Food Science
Author DOMEARA
Credits 5

Level: Advanced

Description of Module / Aims

This module introduces the student to the concerns arising from the increasing globalisation of the food supply chain, and addresses the systems used to ensure food safety and traceability throughout the supply chain, from producer to consumer. In addition, the students will advance their knowledge and skills in genetics and genomics in relation to food traceability and diagnostics. Students will also learn about genetically modified foodstuffs and how they are detected. The module will highlight the regulations within the EU, and will focus on current and emerging methods of food traceability, both electronic and diagnostic (assays and molecular).

Programmes

F00D-0025 BSc (Hons) in Agricultural Science (WD_SAGRI_B)

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Indicative Content

- EU traceability legislation, its current relevance with an emphasis on recent case studies
- Emerging food safety, traceability and security issues
- Traceability systems
- DNA-based diagnostic methodologies such as real-time PCR, DNA sequencing, bioinformatics, single nucleotide polymorphisms (SNPs) and metabarcoding
- Non-molecular based methodologies for food authentication such as chromatography, immunoassays and electronic traceability systems
- Review learning progress using continuous assessment methodologies
- Review and compare relevant literature articles

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Compare the essential components of a food traceability and management system, and discuss the reasons and strategies required for food recall and withdrawal actions.
2. Critique food safety and security issues and their influence on future food production.
3. Evaluate the benefits and disadvantages of genetic manipulation of a plant or animal to agriculture and society.
4. Evaluate the uses and applications of DNA-based diagnostic technologies in relation to food/feed safety and traceability systems.
5. Design and optimise diagnostic assays for food traceability.
6. Compare various methods for use in food authentication and traceability.
7. Appraise the legislative systems governing food/feed traceability (RASFF, TRACES etc.).

Learning and Teaching Methods

- Lectures.

- Laboratory practicals.
- Guest industry speakers.
- Case studies will be employed to reinforce module material.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1, 2, 3, 4, 6, 7
Continuous Assessment	40%	
<i>Lab Report</i>	40%	5

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.

40%-49%: Will be able to show a good knowledge of the key learning outcomes including a good understanding of factors affecting food traceability and the application of various technologies to food traceability.

50%-59%: As well as the above, the student will be able to demonstrate a good understanding of the learning outcomes as well as being able to describe and evaluate the effectiveness of systems used to ensure food safety, traceability and security.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics including legislative systems governing food/feed traceability and demonstrate an ability to relate detailed knowledge acquired to new material.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the ability to fully integrate the various components of the course.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	36	
Independent Learning	75	

Essential Material

- "Food Safety Authority of Ireland." www.fsai.ie
- Alberts, B. , D. Bray, K. Hopkin, A. Johnson, J. Lewis and Martin Raff, Keith Roberts, Peter Walter. *Essential Cell Biology*. USA: Garland, 2013.
- Bertheau, Y. and J. Davison. *Genetically Modified and non-Genetically Modified Food Supply Chains : Co-Existence and Traceability*. UK: John Wiley & Sons, 2013.
- Nollet, L.M.N. and F. Toldre. *Advances in Food Diagnostics*. UK: John Wiley & Sons, 2008.
- Shan, G. *Immunoassays in Agricultural Biotechnology*. UK: John Wiley & Sons, 2011.

Requested Resources

- Room Type: Biology Lab
- Room Type: Computer Lab